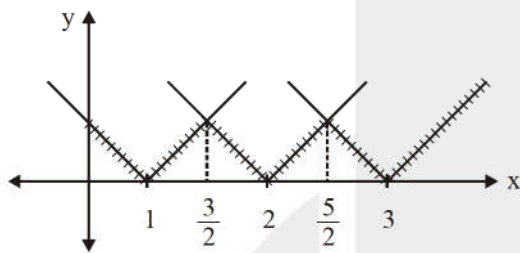


WORK SHEET - 20

INDEFINITE & DEFINITE INTEGRATION

Q.1 (A)

$$\int_0^3 f(x) dx = \frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$$



Q.2 (B)

$$f(x) = \frac{x}{x-1} \Rightarrow f(f(x)) = \frac{\frac{x}{x-1}}{\frac{x}{x-1} - 1} = \frac{x}{x-x+1} = x$$

$\Rightarrow f(f(f(\dots x)))$ when f is odd number of times $= f(x)$

$$\int_{1/6}^{1/3} \frac{x}{x-1} dx = \int_{1/6}^{1/3} \frac{x-1+1}{x-1} = x + \log(x-1) \Big|_{1/6}^{1/3}$$

$$= \left(\frac{1}{3} - \frac{1}{6} \right) + \log \frac{\frac{1}{3} - 1}{\frac{1}{6} - 1} = \frac{1}{6} + \log \frac{4}{5} = \log e^{1/6} \frac{4}{5} \text{ Ans.}$$

Q.3 (B)

$$f(x) = \frac{4x^2 - 4x + 1 + x}{\int_0^{2\pi x} \sin^2 t dt} = \frac{(2x-1)^2}{\int_0^{2\pi x} \sin^4 t dt} + \frac{x}{\int_0^{2\pi x} \sin^4 t dt}$$

$$f'(x) = g'(x) + \frac{\int_0^{2\pi x} \sin^4 t dt \cdot 1 - x \sin^4(2\pi x) \cdot 2\pi}{\left(\int_0^{2\pi x} \sin^4 t dt \right)^2}$$

$$f'\left(\frac{1}{2}\right) = g'\left(\frac{1}{2}\right) + \frac{1}{\int_0^{\pi} \sin^4 t \, dt} = 0 + \frac{1}{2 \cdot \frac{3\pi}{16}} = \frac{8}{3\pi} \Rightarrow p - q = 5$$

Q.4 (C)

$$\int e^{1-\cos x} (1 + x \sin x) dx = e \int e^{-\cos x} (1 + x \sin x) dx$$

$$e \left[\int e^{-\cos x} dx + \int x e^{-\cos x} \cdot \sin x dx \right]$$

Now integrating second integral using integration by parts.

$$e \left[\int e^{-\cos x} dx + x (e^{-\cos x}) - \int 1 \cdot e^{-\cos x} dx \right] = e \cdot x \cdot e^{-\cos x} + c = x \cdot e^{1-\cos x} + c$$

$f(x) = x$ and $g(x) = \cos x$ number of solution is 2.

Q.5 (A)

$$1000 = 5^3 \cdot 2^3$$

$$\text{Number of factors } (3+1)(3+1) = 16$$

$$\text{Applying king, } P = - \int_{-2}^2 x \ln(a_1^{-x} + a_2^{-x} + \dots + a_{16}^{-x}) dx = - \int_{-2}^2 x \ln \left(\frac{1}{a_1^x} + \frac{1}{a_2^x} + \dots + \frac{1}{a_{16}^x} \right) dx$$

$$= - \int_{-2}^2 x \ln \left(\frac{(a_1^x + a_2^x + \dots + a_{16}^x)}{1000^x} \right) dx = - \int_{-2}^2 x \ln(a_1^x + a_2^x + \dots + a_{16}^x) dx + \int_{-2}^2 x^2 \ln(1000) dx$$

$$2P = \frac{16}{3} \ln 1000$$

$$P = 8 \ln 10 \Rightarrow \frac{2P}{n} = \frac{16 \ln 10}{16} = \ln 10 \text{ Ans.}$$

Q.6 (C)

$$\text{We have } f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dt = \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} t f(t) dt$$

$$\therefore f(x) = (\pi + 1) \sin x + A \quad \dots (1)$$

$$\text{Now } A = \int_{-\pi/2}^{\pi/2} t ((\pi + 1) \sin t + A) dt = 2(\pi + 1) \int_0^{\pi/2} t \sin t dt \quad \text{(By part)}$$

$$\Rightarrow A = 2(\pi + 1)$$

$$\text{Hence } f(x) = (\pi + 1) \sin x + 2(\pi + 1)$$

$$\therefore f_{\max} = 3(\pi + 1) = M$$

$$\text{and } f_{\min} = (\pi + 1) = m \Rightarrow \frac{M}{m} = 3$$

Q.7 $\int \frac{x - \sin x \cos x}{x^2 \cos^2 x} dx = F(x)$ such that $F(1) = 1$, then $\lim_{x \rightarrow 0} F(x)$ equals

- (A) 1 (B) $1 - \tan 1$ (C*) $2 - \tan 1$ (D) $\tan 1$

$$\int \frac{x - \sin x \cos x}{x^2 \cos^2 x} dx = \int \frac{dx}{x \cos^2 x} - \int \frac{\tan x}{x^2} dx$$

$$\frac{1}{x} \tan x + \int \frac{1}{x^2} \tan x dx - \int \frac{\tan x}{x^2} dx$$

$$F(x) = \frac{1}{x} \tan x + c; F(1) = 1 \Rightarrow c = 1 - \tan 1$$

$$\therefore F(x) = \frac{\tan x}{x} + 1 - \tan 1 \text{ Ans.}$$

Q.8 (B)

Q.9 (C)

Q.10 (D)

$$(1 + x^2) f(x) \cdot f\left(\frac{2}{x}\right) + x = x f(x) + (1 + x^2) f\left(\frac{2}{x}\right)$$

$$(1 + x^2) f\left(\frac{2}{x}\right) (f(x) - 1) = x (f(x) - 1)$$

$$(f(x) - 1) \cdot \left((1 + x^2) f\left(\frac{2}{x}\right) - x \right) = 0$$

$$f(x) = 1 \text{ (rejected) and } f\left(\frac{2}{x}\right) = \frac{x}{1 + x^2}$$

$$f\left(\frac{2}{x}\right) = \frac{x}{1 + x^2}, x \neq 0$$

$$f(t) = \frac{2t}{4 + t^2}, f(0) = 0 \quad \{\because f \text{ is continuous}\}$$

(i) $\lim_{x \rightarrow 0} \left(\tan^{-1} \left(\frac{2x}{4 + x^2} \right) + a \right)^{\frac{4}{x}} = e^p$

a must be 1.

$$\Rightarrow e^{\lim_{x \rightarrow 0} \left(\tan^{-1} \left(\frac{2x}{4 + x^2} \right) \right) \cdot \frac{4}{x}} = e^p$$

$$\Rightarrow p = \lim_{x \rightarrow 0} \frac{\tan^{-1} \left(\frac{2x}{4 + x^2} \right)}{x \cdot \frac{2}{4 + x^2}} \cdot \frac{4 \cdot 2}{4 + x^2} = 2 \text{ Ans.}$$

$$(ii) \quad I = \int_0^{\infty} \frac{\ln(2f(x))}{4+x^2} dx = \int_0^{\infty} \frac{\ln\left(\frac{2 \cdot 2x}{4+x^2}\right)}{4+x^2} dx$$

$$\text{Put } x = 2 \tan \theta \Rightarrow dx = 2 \sec^2 \theta \cdot d\theta$$

$$= \int_0^{\pi/2} \frac{\ln\left(\frac{4 \cdot 2 \tan \theta}{4+4 \tan^2 \theta}\right)}{4+4 \tan^2 \theta} \cdot 2 \sec^2 \theta \cdot d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \ln(\sin 2\theta) d\theta = \frac{1}{2} \left(\frac{-\pi}{2} \ln 2 \right) = \frac{-\pi}{4} \ln 2 \quad \text{Ans.}$$

$$(iii) \quad \int_0^2 g(x) dx + \int_0^{1/2} g^{-1}(x) dx = 1$$

$$\Rightarrow \int_0^{1/2} g^{-1}(x) dx = 1 - \int_0^2 \frac{2x}{4+x^2} dx = 1 - \left(\ln(4+x^2) \right)_0^2 = 1 - (\ln 8 - \ln 4) = 1 - \ln 2 \quad \text{Ans.}$$

Q.11 (ABCD)

$$I_1 = \int_0^{\pi} \frac{\cos x}{(x+2)^2} dx, \quad x = 2t$$

$$I_1 = \int_0^{\pi/2} \frac{\cos 2t}{2(t+1)^2} dt = \frac{-\cos 2t}{2(t+1)} \Big|_0^{\pi/2} + \int_0^{\pi/2} \frac{2 \sin 2t}{2(t+1)} dt$$

$$I_1 = \frac{1}{2+\pi} + \frac{1}{2} - \int_0^{\pi/2} \frac{4 \sin t \cos t}{2(t+1)} dt$$

$$2I_1 = \frac{2}{2+\pi} + 1 - 4I_2$$

$$4I_2 = \frac{2}{2+\pi} + 1 - 2I_1 \quad \text{Ans.}$$

Q.12 (AB)

$$f(x) = \int_0^x (f(t))^{-1} dt \Rightarrow f'(x) = \frac{1}{f(x)} \cdot 1$$

$$f'(x) \cdot f(x) = 1$$

$$\frac{(f(x))^2}{2} = x + c \Rightarrow f(x) = \sqrt{2(x+c)}$$

$$\text{and } f(1) = \sqrt{2} \Rightarrow c = 0 \quad \text{i.e. } f(x) = \sqrt{2x} \quad \text{Ans.}$$

Q.13 (CD)

$$g \circ f(x) = \sec^{-1} \sqrt{1 - \frac{1}{x}}$$

$$\text{Domain: } \sqrt{1 - \frac{1}{x}} \geq 1 \Rightarrow x < 0 \Rightarrow x \in (-\infty, 0)$$

$$\text{Range: } \left(0, \frac{\pi}{2}\right)$$

$$\cot(g(f(x))) = \sqrt{-x} \Rightarrow \int_{-4}^{-1} \sqrt{-x} \, dx = \frac{14}{3}$$

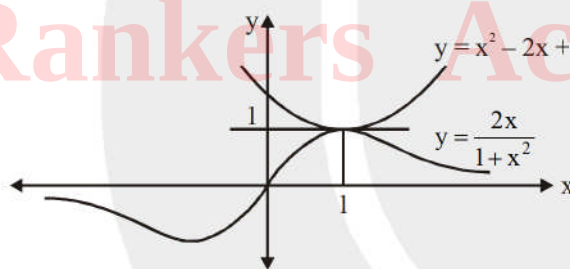
$$\lim_{x \rightarrow -\infty} \frac{\sqrt{-x}}{\sqrt{1-x} + \sqrt{2-x}} = \frac{1}{2}$$

Q.14 (ABCD)

Differentiating given relation

$$f(x) = x - x^2 f(x) \Rightarrow f(x) = \frac{x}{x^2 + 1}$$

$$\therefore \text{Given equation becomes } x^2 - 2x + 2 = \frac{2x}{1+x^2}$$



The given equation is $x^2 - 2x + 2 = 2f(x)$
 $\Rightarrow$ only one possible solution

Ans. (A), (B), (C), (D)

Q.15 (A) P; (B) S; (C) T; (D) Q

$$\int_1^{x^2} \frac{d}{dx} \left(\frac{3x^2 + 1}{x^7 + 2x^5 + 2x^4 + x^3 + 2x^2 + 5x} \right) dx$$

$$x^2 \cdot \frac{3x^2 + 1}{x^7 + 2x^5 + 2x^4 + x^3 + 2x^2 + 5x} - \int \frac{2x(3x^2 + 1) dx}{x(x^6 + 2x^4 + 2x^3 + x^2 + 2x + 5)}$$

$$\frac{x(3x^2 + 1)}{x^6 + 2x^4 + 2x^3 + x^2 + 2x + 5} - \int \frac{2(3x^2 + 1) dx}{(x^3)^2 + 2x^3(x+1) + (x+1)^2 + 4}$$

$$\frac{x(3x^2+1)}{x^6+2x^4+2x^3+x^2+2x+5} - 2 \int \frac{(3x^2+1)dx}{(x^3+(x+1))^2+4}$$

$$\frac{x(3x^2+1)}{x^6+2x^4+2x^3+x^2+2x+5} - \tan^{-1}\left(\frac{x^3+x+1}{2}\right) + c$$

$$g(x) = \frac{x^3+x+1}{2}$$

(A) $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}(1) \quad \left\{ \because g(0) = \frac{1}{2} \right\}$

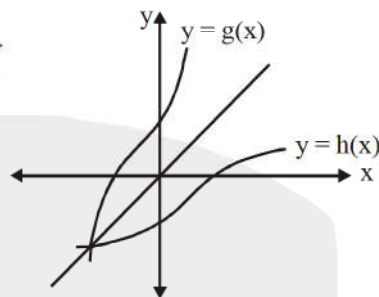
(B) $g(x) = h(x)$ has exactly one solution

$$\frac{q}{\pi} \sin^{-1}\left(\frac{1}{2}\right) = 1 \Rightarrow q = 6$$

(C) $h'\left(\frac{31}{2}\right) = \frac{1}{g'(3)} = \frac{1}{14} = \frac{g(0)}{k} \Rightarrow k = 7$

(D) $\int_{-3}^3 g(x) dx = \int_{-3}^3 \left(\frac{x^3+x+1}{2}\right) dx = \frac{1}{2} \cdot 2 \int_0^3 dx = 3 = 2g(1)$

$$\Rightarrow n = 1 \text{ Ans.}$$



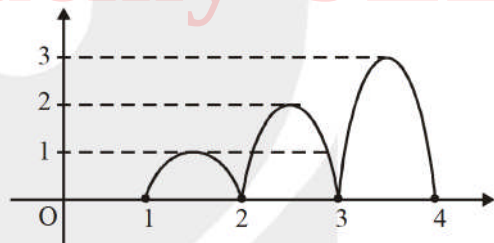
Q.16 [5050]

Let $y = f(x) = [x] \sin \pi x$

$$A = \int_0^1 0 dx + \int_1^2 -\sin \pi x dx + \int_2^3 2 \sin \pi x dx + \dots$$

$$A = \frac{2}{\pi} + \frac{4}{\pi} + \frac{6}{\pi} + \dots + \frac{200}{\pi}$$

$$A = \frac{2}{\pi} (1 + 2 + \dots + 100) \Rightarrow \frac{A\pi}{2} = 5050 \text{ Ans.}$$



Q.17 [6]

$$f''(t) = \int_{12t}^{t^3+6t^2} \ln \left| \frac{x+8}{(t+2)^3-x} \right| dx \quad \dots (i)$$

Applying king

$$f''(t) = \int_{12t}^{t^3+6t^2} \ln \left| \frac{t^3+6t^2+12t-x+8}{(t+2)^3-(t^3+6t^2+12t-x)} \right| dx$$

$$f''(t) = \int_{12t}^{t^3+6t^2} \ln \left| \frac{(t+2)^3-x}{8+x} \right| dx \quad \dots (ii)$$

(i) + (ii)

$$2f''(t) = 0$$

$$\Rightarrow f''(t) = 0$$

$$\Rightarrow f'(t) = \lambda$$

$$f(t) = \lambda t + \mu$$

$$\left. \begin{aligned} f(1) = 2 &\Rightarrow \lambda + \mu = 2 \\ f(2) = 1 &\Rightarrow 2\lambda + \mu = 1 \end{aligned} \right\} \Rightarrow \lambda = -1, \mu = 3$$

$$\therefore \int_{-1}^1 (-x + 3) dx = 2 \int_0^1 3 \cdot dx = 6 \text{ Ans.}$$

Q.18 [0003]

$$\frac{\pi}{2} \int \frac{dx}{(\cos^{-1} x)^2 \sqrt{1-x^2}}$$

$$\text{Let } \cos^{-1} x = t \Rightarrow \frac{dx}{\sqrt{1-x^2}} = -dt$$

$$\frac{-\pi}{2} \int \frac{dt}{t^2} = \left(\frac{\pi}{2}\right) \cdot \frac{1}{t} + c = \frac{\sin^{-1} x + \cos^{-1} x}{\cos^{-1} x} + c = \left(\frac{\sin^{-1} x}{\cos^{-1} x}\right)_0^{1/2} = \left(\frac{\pi/6}{\pi/3}\right) = \frac{3}{6} = \frac{1}{2} \text{ Ans.}$$

Q.19 [7]

$$\int e^x \left(\frac{1-x^n}{1-x} \right) dx = e^x P(x) + C$$

$$\text{Let } P(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_{n-1} x^{n-1}$$

$$= \int e^x (1 + x + x^2 + \dots + x^{n-1}) dx = e^x (a_0 + a_1 x + a_2 x^2 + \dots + a_{n-1} x^{n-1}) + c$$

$$P(0) = a_0 = 620$$

Differentiating both sides

$$e^x (1 + x + x^2 + \dots + x^{n-1}) = e^x (a_1 + 2a_2 x + 3a_3 x^2 + \dots + (n-1)a_{n-1} x^{n-2}) + (a_0 + a_1 x + a_2 x^2 + \dots + a_{n-1} x^{n-1}) e^x$$

Comparing coefficient of same power of x

$$a_{n-1} = 1$$

$$a_0 = 620$$

$$a_1 + a_0 = 1 \Rightarrow a_1 = -619$$

$$a_1 + 2a_2 = 1 \Rightarrow a_2 = 310$$

$$a_2 + 3a_3 = 1 \Rightarrow a_3 = -103$$

$$a_3 + 4a_4 = 1 \Rightarrow a_4 = +26$$

$$a_4 + 5a_5 = 1 \Rightarrow a_5 = -5$$

$$a_5 + 6a_6 = 1 \Rightarrow a_6 = 1$$

$$\therefore n-1 = 6 \Rightarrow n = 7.$$